Mayank Kapadi

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Professional Summary

Data-driven engineer with hands-on experience building hybrid physics ML models, advanced time-series prediction systems, and analytical solutions for real-world problems. Skilled in Python, PyTorch, scientific computing, and data processing, with research experience combining ODEs and neural networks for dynamic system modeling. Strong foundation in machine learning, numerical methods, and system simulation, supported by industry experience in software development and data-centric applications. Passionate about developing reliable, scalable ML solutions for complex technical domains.

Experience

Student Research Assistant,University of Paderborn, Paderborn, Germany — Apr 2025 – Present

- Conducting research on hybrid modeling techniques combining **ODEs and ANNs** for solving inverse dynamics problems.
- Developing and training **physics-informed ML models** for time-series prediction in mechanical systems.
- Implementing experiments using Python, PyTorch, NumPy, SciPy, and visualization tools such as Matplotlib.
- Collaborating with research teams to analyze system behavior and evaluate predictive performance.

Student Research Assistant, University of Paderborn, Paderborn, Germany — Mar 2022 – Apr 2024

- Contributed to the development of an educational **Unity-based teaching platform** for interactive learning.
- Designed and implemented UI components and interactive modules using C, Unity3D, and Git.
- Integrated backend elements using HTML, PHP, CSS, and SQL to support platform functionality.
- Worked closely with educators to translate instructional needs into functional software features.

Software Developer, Ashwamedh Infotech Pvt. Ltd., Mumbai, India — Jun 2018 – Dec 2020

- Integrated database management systems with payroll processing platforms for multiple enterprise clients.
- Diagnosed and resolved complex client-side issues, documenting solutions and QA procedures thoroughly.
- Ensured robust code quality through systematic testing and validation of newly developed features.
- Delivered customized software solutions using HTML, PHP, CSS, SQL, and JavaScript.
- Provided technical support and problem resolution for high-impact system issues.

Education

M.Sc. Computer Science, University of Paderborn, Germany — Apr 2021 – Sep 2025

B.E. Computer Engineering, University of Mumbai, India — Jun 2014 – May 2018

Skills

Programming & ML: Python, PyTorch, NumPy, SciPy, Pandas, SQL, PHP, HTML, CSS, JavaScript, C#, LaTeX
Tools & Platforms: Git, Unity3D, Blender
Other: Version Control, Documentation, Problem-Solving

Languages

English (C1 - Business fluency)
German (B1 - Currently learning)

Projects

- **Software Architecture Concept for Vehicles (Master's Thesis, Fraunhofer IEM):** Developed a software architecture concept for software-intensive vehicles, integrating advanced sensors, V2X communication, security mechanisms, and AI/ML considerations. Created complete SysML models in Cameo and built a functional MATLAB/Simulink simulation.
- **Adaptive Virtual Chemistry Laboratory (VIRTUCHEMLAB):** Implemented a collaborative VR-based digital twin of a chemistry laboratory, including interaction logic, UI components, and 3D environment design using Unity3D, Blender, C#, Python, and Git.
- **Smart Grid Technology – Monitoring & Prediction (B.E. Project):** Designed a smart grid system to monitor household electricity usage and predict future consumption using algorithmic forecasting methods.

Certifications

AWS Certified Cloud Practitioner (Credly)
Microsoft MTA Security fundamentals certified (Credly)